Key Concepts

Absolute Value Equation $|x| = a \Rightarrow x = a$ or $x = -a$

Absolute Value Inequality ($\leq$) $|x| \leq a \Rightarrow -a \leq x \leq a$ (AND, closed interval)

Absolute Value Inequality ($\geq$) $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a$ (OR, union)

Verbal meaning: $|x - c| \leq d$ means "the distance between x and c is at most d."

Steps to Solve:

1. Isolate the absolute value expression.
2. Split into two cases (positive and negative).
3. Solve each case and check your answers.
4. Graph the solution on a number line.

Example 1 — Equation: $|f - 6| = 3$

Step 1: Split into two cases

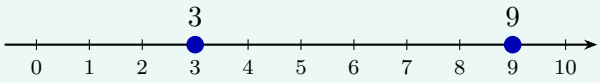
$$f - 6 = 3 \quad \text{or} \quad f - 6 = -3$$

Step 2: Solve each

$$f = 9 \quad \text{or} \quad f = 3$$

Step 3: Check $f = 9$: $|9 - 6| = 3$ ✓ $f = 3$: $|3 - 6| = 3$ ✓

Solution: $f = 3$ or $f = 9$

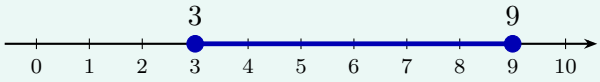


Example 2 — Inequality: $|f - 6| \leq 3$

Method (Algebraic): Apply $-3 \leq f - 6 \leq 3$, then add 6:

$$3 \leq f \leq 9$$

Check: $f = 4$: $|4 - 6| = 2 \leq 3$ ✓ $f = 10$: $|10 - 6| = 4 \leq 3$ × (non-solution, correct)



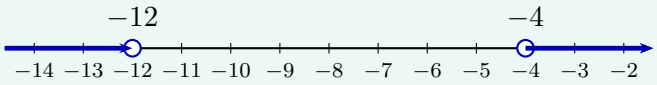
Example 3 — Inequality: $5|b + 8| - 7 > 13$

Step 1: Isolate $5|b + 8| > 20 \Rightarrow |b + 8| > 4$

Step 2: Split (OR)

$$b + 8 < -4 \quad \text{or} \quad b + 8 > 4$$

$$b < -12 \quad \text{or} \quad b > -4$$

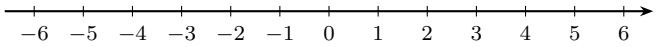


Practice Problems

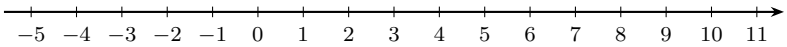
Solve each equation or inequality. Show all steps. Graph your solution on the number line provided.

Section A — Absolute Value Equations

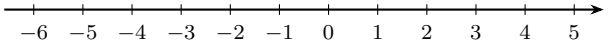
1. $|x| = 5$



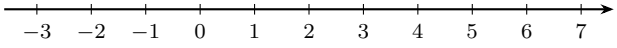
2. $|x - 3| = 7$



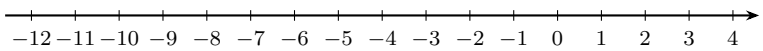
3. $|2x + 1| = 9$



4. $|3x - 6| = 12$

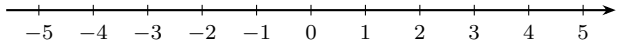


5. $|x + 4| - 2 = 5$

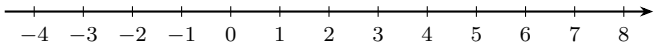

−12 −11 −10 −9 −8 −7 −6 −5 −4 −3 −2 −1 0 1 2 3 4

Section B — Inequalities ($\leq$ / $<$) "AND" type ⇒ closed interval

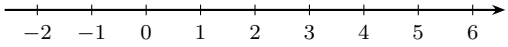
6. $|x| \leq 4$



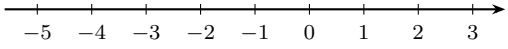
7. $|x - 2| < 5$



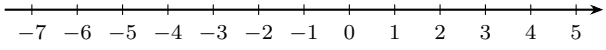
8. $|2x - 4| \leq 6$



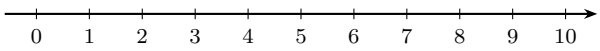
9. $|3x + 3| < 9$



10. $2|x + 1| - 3 \leq 7$

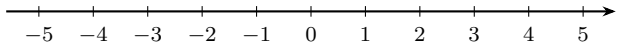


11. $3|x-5|+1 \leq 10$

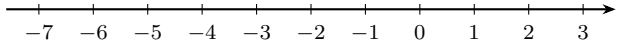


Section C — Inequalities ($\geq$ / $>$) "OR" type ⇒ union of two rays

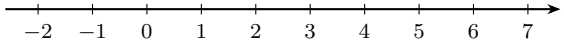
12. $|x| > 3$



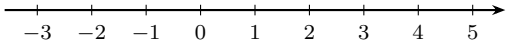
13. $|x + 2| \geq 4$



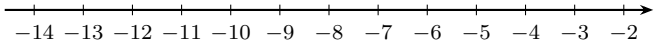
14. $|2x - 6| > 4$



15. $3|x - 1| \geq 9$

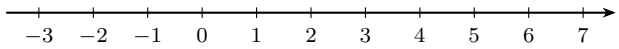


16. $5|b + 8| - 7 > 13$

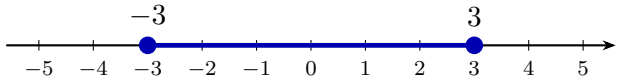


Section D — Mixed & Word Problems

17. Solve and graph: $|4x - 8| \geq 12$



18. Write the absolute value inequality shown by the graph below, then verify with one solution and one non-solution.



Inequality: _____

---

19. Temperature: A thermostat keeps the temperature within 3 degrees of 70°F. Write and solve an absolute value inequality to find the acceptable temperature range.

---

20. Challenge: Solve $|2x - 1| = |x + 4|$. *Hint: The expressions inside can be equal, or one can be the negative of the other.*

"The distance interpretation of absolute value makes everything click. Keep practicing!"

Answer Key

Section A — Absolute Value Equations

1. $|x| = 5$ — $\boldsymbol{x = -5}$ **or** $\boldsymbol{x = 5}$
2. $|x - 3| = 7$ — $\boldsymbol{x = -4}$ **or** $\boldsymbol{x = 10}$
3. $|2x + 1| = 9$ — $\boldsymbol{x = -5}$ **or** $\boldsymbol{x = 4}$
4. $|3x - 6| = 12$ — $\boldsymbol{x = -2}$ **or** $\boldsymbol{x = 6}$
5. $|x + 4| - 2 = 5 \Rightarrow |x + 4| = 7$ — $\boldsymbol{x = -11}$ **or** $\boldsymbol{x = 3}$

Section B — Inequalities ($\leq$ / $<$)

6. $|x| \leq 4$ — $\boldsymbol{-4 \leq x \leq 4}$
7. $|x - 2| < 5 \Rightarrow -5 < x - 2 < 5$ — $\boldsymbol{-3 < x < 7}$
8. $|2x - 4| \leq 6 \Rightarrow -6 \leq 2x - 4 \leq 6$ — $\boldsymbol{-1 \leq x \leq 5}$
9. $|3x + 3| < 9 \Rightarrow -9 < 3x + 3 < 9$ — $\boldsymbol{-4 < x < 2}$
10. $2|x + 1| - 3 \leq 7 \Rightarrow |x + 1| \leq 5$ — $\boldsymbol{-6 \leq x \leq 4}$
11. $3|x - 5| + 1 \leq 10 \Rightarrow |x - 5| \leq 3$ — $\boldsymbol{2 \leq x \leq 8}$

Section C — Inequalities ($\geq$ / $>$)

12. $|x| > 3$ — $\boldsymbol{x < -3}$ **or** $\boldsymbol{x > 3}$
13. $|x + 2| \geq 4 \Rightarrow x + 2 \leq -4$ or $x + 2 \geq 4$ — $\boldsymbol{x \leq -6}$ **or** $\boldsymbol{x \geq 2}$
14. $|2x - 6| > 4 \Rightarrow 2x - 6 < -4$ or $2x - 6 > 4$ — $\boldsymbol{x < 1}$ **or** $\boldsymbol{x > 5}$
15. $3|x - 1| \geq 9 \Rightarrow |x - 1| \geq 3$ — $\boldsymbol{x \leq -2}$ **or** $\boldsymbol{x \geq 4}$
16. $5|b + 8| - 7 > 13 \Rightarrow |b + 8| > 4$ — $\boldsymbol{b < -12}$ **or** $\boldsymbol{b > -4}$

Section D — Mixed & Word Problems

17. $|4x - 8| \geq 12$
    $4x - 8 \leq -12 \Rightarrow x \leq -1$ or $4x - 8 \geq 12 \Rightarrow x \geq 5$ **$x \leq -1$ or $x \geq 5$**

18. Graph shows closed dots at -3 and 3, shaded between.
    Inequality: $|x| \leq 3$ Solution: **$-3 \leq x \leq 3$**
    Check: $x = 0$: $|0| = 0 \leq 3$ ✓ $x = 4$: $|4| = 4 \leq 3$ ×

19. $|T - 70| \leq 3 \Rightarrow -3 \leq T - 70 \leq 3$ **$67°\text{F} \leq T \leq 73°\text{F}$**

20. $|2x - 1| = |x + 4|$
    Case 1: $2x - 1 = x + 4 \Rightarrow x = 5$
    Case 2: $2x - 1 = -(x + 4) \Rightarrow 3x = -3 \Rightarrow x = -1$ **$x = -1$ or $x = 5$**